MATH 20250 HW5: KEYS AND HINTS

RUOHAN HU

Contents

1. General Advice

Check your answer every time you feel like doing so. There are easy way to check your computations.

For matrix multiplication, try to check if the determinants are correct (only when the determinants are easy to compute, of course). Use the multiplicativity of determinant.

For matrix similarity, try to check if the traces are correct. Use the fact that traces (and also determinants) are the same for similar matrices.

For eigenvectors, check if it is really an eigenvector, of course.

2. Q4.1.1

The answer is $FTTFTFFT$.

(b) actually depends on whether the base field is finite. For LADW problems we consider only $\mathbb{R}$ or $\mathbb{C}$ as base fields.

(f) should be false. Consider any diagonalizable matrix. It is similar to the diagonal matrix. The eigenvectors of the diagonal matrix are the standard basis of $\mathbb{R}^n$. The same is not necessarily true for any diagonalizable matrix.

Date: Spring 2023.

3. Q4.1.3

The eigenvectors are $v_1 = (i, 1)$, $v_2 = (-i, 1)$, up to multiplication of a non-zero scalar.

The corresponding eigenvalues are $\lambda_1 = \cos\alpha + i\sin\alpha$ and $\lambda_2 = \cos\alpha - i\sin\alpha$.

4. Q4.1.6

Some students believe that 0 is an eigenvector corresponding to the eigenvalue 0 in this problem.

Actually, 0 cannot be an eigenvector because it is stated in the definition that an eigenvector is a non-zero vector.

We have proved that if a matrix is nilpotent, then it is non-invertible. Because the matrix is non-invertible, its kernel (null space) must have dimension at least 1. A basis of the null space is an eigenvector corresponding to 0.

5. Q4.2.1

The answer is TFT.

6. Q4.2.3

One possible diagonalization is

$$\begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

The result is

$$\begin{pmatrix} \frac{3}{4}5^{2004} + \frac{1}{4} & \frac{3}{4}5^{2004} - \frac{3}{4} \\ \frac{1}{4}5^{2004} - \frac{1}{4} & \frac{1}{4}5^{2004} + \frac{3}{4} \end{pmatrix}$$

7. Q4.2.4

The answer is

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix}.$$

The matrix is indeed unique. One reason to argue this is that a matrix is determined by its action on a basis (of $\mathbb{R}^n$). $[1, 1]^T$ and $[1, 2]^T$ are indeed a basis, and the matrix's action on them is determined. Thus the matrix is unique.

Also, if we switch the diagonal entries, and switch the order of the columns of the transformation matrices accordingly, we would still get the same matrix.

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix}.$$

8. Q4.2.6

The three eigenvectors are $(1, 0, 0)^T$, $(2, 1, 0)^T$, $(3, 2, 1)^T$. The corresponding eigenvalues are $2, 5, 4$.

9. Q4.2.7

The three eigenvectors are $(1, 0, 0)^T$, $(0, 1, 0)^T$, $(3, 2, 1)^T$. The corresponding eigenvalues are $2, 2, 4$.

10. Q4.2.8

To calculate the possible B such that $B^2 = A$, we first diagonalize A as $A = PDP^{-1}$, where C is the diagonal matrix of A, which is going to be

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}.$$

(Or maybe $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. This would not change the final result, since the transformation matrix would also be different.)

Then, we need only to find the square root of D. Suppose we have D' such that $(D')^2 = D$. Then, $B = PD'P^{-1}$ would automatically be the square root of A.

Now, there are going to be four possibilities of D':

$$D' = \begin{pmatrix} \pm\sqrt{3} & 0 \\ 0 & \pm\sqrt{2} \end{pmatrix}$$

The sign of the two entries can be different, so there are four possibilities. Thus, there are four possibilities for B.

11. Q4.2.10

Sketch of a proof goes as follows: the matrix can have at most 5 eigenvalues with multiplicity. Because it has one three-dimensional eigenspace, one of its eigenvalue must have multiplicity 3. Thus, because it has two other eigenvalues, its total number of eigenvalues counting multiplicity is exactly 5. Thus it is diagonalizable.

12. Q4.2.11

I would like to collect a menagerie of diagonalizable matrices that some students believe to be non-diagonalizable.

The matrix

$$\begin{pmatrix} 2 & & 3 \\ & 2 & \\ & & 1 \end{pmatrix}$$

is diagonalizable. Its three eigenvalues are $2, 2, 1$. The three corresponding eigenvectors are $[1, 0, 0]^T$, $[0, 1, 0]^T$, $[3, 0, -1]^T$.

The matrix

$$\begin{pmatrix} 1 & 1 & \\ & 2 & \\ & & 3 \end{pmatrix}$$

is diagonalizable. Its three eigenvalues are $1, 2, 3$. The corresponding eigenvectors are $[1, 0, 0]^T$, $[1, 1, 0]^T$, $[0, 0, 1]^T$.

The non-diagonalizable cases are, for example, matrices of the form $(a \neq 0)$

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$$

13. Supplementary Problem 1(b)

A misconception here is to think that "a diagonalizable matrix must be invertible." Many diagonal matrices themselves, for example, are not invertible, such as the diagonal matrices with 0 as some of its diagonal entries.

One example of a matrix that is non-invertible but still diagonalizable is

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}.$$

The reason behind such misconception is probably that the student mistakenly believes eigenvalues cannot be zero. While eigenvectors cannot be zero, **eigenvalues can be zero**.

A good example of non-diagonal matrix would be

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

It can be proved that its eigenvectors are all scalar multiples of $(1, 0)$, so there is not a basis of eigenvectors.